

Arabic Pronunciation Assessment System Using Speech Signal Processing

Raghad Jamhour, Woroud Amjad

Department of Electrical and Computer Engineering, Birzeit University, Palestine

1220212@student.birzit.edu, 1220357@student.birzit.edu

Abstract

This project presents an automatic Arabic Pronunciation Assessment System designed to evaluate and provide feedback on spoken Arabic using speech signal processing techniques. The motivation stems from the difficulty non-native learners face in correctly producing Arabic phonemes, particularly emphatic, pharyngeal, and uvular sounds, which are not present in many other languages.

The system compares a reference pronunciation with a test utterance and generates an overall pronunciation score along with feature-level feedback. The methodology includes audio preprocessing (silence removal, normalization, framing), followed by acoustic feature extraction. The extracted features include Mel-Frequency Cepstral Coefficients (MFCCs), pitch, duration, formants, zero-crossing rate, and spectral features. Formant analysis is performed using Praat, while the remaining features are computed using Python-based signal processing tools.

Similarity between reference and test signals is calculated using cosine similarity for MFCC features and normalized distance-based metrics for other features. A weighted scoring mechanism combines all feature scores to produce a final pronunciation score between 0% and 100%. Experiments were conducted on a dataset of 10 Arabic words spoken by 5 speakers, including a controlled mispronunciation test case.

Results show that the system produces consistent scores across speakers and successfully detects pronunciation deviations, with lower scores for intentionally mispronounced inputs. The system demonstrates that classical speech signal processing can effectively support lightweight Arabic pronunciation assessment without requiring deep learning models.

1. Introduction

Arabic is one of the most widely spoken languages in the world, yet it presents significant challenges for non-native learners due to its unique phonetic inventory. Sounds such as emphatic consonants, pharyngeal fricatives, and uvular consonants have no equivalent in most other languages, making correct pronunciation particularly difficult to acquire. Traditional language learning methods rely heavily on human instructors to evaluate pronunciation quality, which is both time-consuming and subjective.

Automatic pronunciation assessment systems offer a promising solution by providing learners with objective, consistent, and immediate feedback on their pronunciation. Such systems leverage speech signal processing techniques to analyze and compare spoken recordings against reference recordings, enabling scalable and accessible language learning tools.

In this project, we design and implement an Arabic Pronunciation Assessment System. The system takes a reference pronunciation and a student pronunciation as input, and produces an overall pronunciation score along with detailed feedback. The pipeline consists of several stages: speech recording and preprocessing, short-time signal analysis, feature extraction, pronunciation comparison, scoring, and feedback generation. The system focuses on a set of ten Arabic words carefully chosen to cover the most phonetically challenging sounds in the Arabic language.

2. Background

Automatic pronunciation assessment has been an active area of research in spoken language processing. Early work by Eskenazi laid the groundwork for spoken language technology in educational contexts, establishing the importance of computer-assisted pronunciation teaching (CAPT) systems. Since then, numerous studies have explored different approaches to pronunciation evaluation across various languages.

For feature extraction, Mel-Frequency Cepstral Coefficients (MFCCs) have remained the dominant technique since their introduction by Davis and Mermelstein in 1980. MFCCs capture the spectral envelope of speech in a compact and perceptually meaningful representation, making them highly effective for characterizing phonetic content across speakers. Arabic pronunciation assessment specifically presents unique challenges due to the language's rich phonetic inventory. A comprehensive review of existing Arabic CAPT systems highlights that most work focuses on three tasks: pronunciation rating, mispronunciation detection and diagnosis, and feedback generation. Research using Hidden Markov Model based speech recognizers has demonstrated the feasibility of automatic Arabic pronunciation scoring, though Arabic remains significantly under-resourced compared to languages like English and Mandarin. More recently, deep learning approaches have been applied to Arabic phoneme detection, with one study achieving a classification accuracy of 95.77% on Arabic short vowels using neural networks.

For pronunciation comparison, distance-based methods such as Euclidean distance and cosine similarity on MFCC feature vectors have been widely used as lightweight and interpretable baselines before applying more complex machine learning models. Formant analysis using tools such as Praat has also been established as a standard method for characterizing vowel quality and place of articulation in phonetic research. In this project, drawing on established methods including MFCCs, pitch extraction, formant analysis, and distance-based comparison to build a lightweight Arabic pronunciation assessment system without relying on large pre-trained models.

3. Methodology

3.1. Dataset and Data Collection

The speech dataset was collected specifically for this project and consists of 50 recordings in total, covering 10 Arabic words spoken by 5 different speakers. The 10 words were carefully selected to represent the most phonetically challenging sounds in the Arabic language, spanning four phonetic categories: emphatic consonants (ض, ط, ظ), pharyngeal fricatives (ع, ح), uvular consonants (ق, غ, خ), and regular consonants (ب, ت). The full list of words is presented in Table 1.

Recordings were captured using the Voice Memos application on a mobile device, with each speaker pronouncing each word once in a quiet environment. Raw recordings were saved in M4A format and subsequently converted to WAV format using the online tool available at online-audio-converter.com, with the following standardized parameters: mono channel, 16,000 Hz sample rate. Each file was named using the convention `word_speakernumber.wav` (e.g. `arabi_1.wav`, `arabi_2.wav`) to ensure consistent organization across the dataset.

Word	Arabic	Sound Category
arabi	عربي	Pharyngeal (ع)
hadeka	حديقة	Pharyngeal (ح)
qalam	قلم	Uvular (ق)
daw	ضوء	Emphatic (ض)
ghurfa	غرفة	Uvular (غ)
khalid	خالد	Uvular (خ)
sadeeq	صديق	Emphatic (ص)
tareeq	طريق	Emphatic (ط)
zarf	ظرف	Emphatic (ظ)
thalatha	ثلاثة	Regular (ث)

Table 1: List of selected Arabic words and phonetic categories

3.2. Preprocessing

Before feature extraction and model training, additional preprocessing was applied to improve signal quality and prepare the data for acoustic analysis. Since the recordings were already standardized to mono and a 16 kHz sampling rate during conversion, no further channel conversion or resampling was performed at this stage.

Silence removal was applied using an energy-based trimming method implemented via `librosa.effects.trim`. Low-energy segments at the beginning and end of each recording were removed using a threshold of -30 dB to reduce non-speech content and isolate the spoken recordings.

After trimming, amplitude normalization was performed by scaling each signal by its maximum absolute value, ensuring a consistent amplitude range across all recordings and reducing variability due to recording conditions.

The cleaned signals were stored in a separate directory (`clean_data`) for further processing. Each signal was then segmented into overlapping frames using a frame length of 512 samples (≈ 32 ms at 16 kHz) and a hop length of 256 samples, resulting in a 50% overlap between consecutive frames. A Hann window was applied to each frame to reduce spectral leakage at frame boundaries.

Finally, waveform visualizations were generated for qualitative inspection of preprocessing effectiveness.

3.3. Signal Analysis

3.3.1. Overview of Acoustic Representations

After preprocessing, the cleaned speech recordings were analyzed using several acoustic representations to examine Arabic pronunciation patterns across different phonetic categories, particularly pharyngeal, uvular, and emphatic consonants. The analysis was performed across all 10 selected words in the dataset (10 words $\times$ 5 speakers), aiming to characterize speech signals using complementary time-domain, frequency-domain, and articulatory features.

The acoustic representations used in this study include waveform, short-time energy, zero-crossing rate, spectrogram, pitch contour, and formant analysis. These features provide a multi-level description of speech, capturing amplitude variations, spectral content, and vocal tract resonances. Formant analysis was conducted using Praat for all 10 words to extract detailed measurements of speech articulation. Representative Praat visualizations for selected words are included to illustrate typical acoustic patterns.

3.3.2. Time-Domain Analysis (waveform, STE, ZCR)

Time-domain analysis was used to examine how the amplitude of the speech signal changes over time. The waveform provides a direct view of the speech signal and helps identify the duration, pauses, and intensity variation of each recording. Short-time energy was calculated for each frame to measure the strength of the speech signal over time. This is useful for locating high-energy regions, stressed syllables, and strong consonant or vowel productions. Zero-crossing rate was also calculated to measure how often the signal crosses the zero-amplitude axis. Higher zero-crossing values are typically associated with noisy or unvoiced speech segments, such as fricatives.

In this project, time-domain analysis was used to observe differences between Arabic sounds with different articulation types. For instance, the waveform of *arabi* after preprocessing (Figure 3.1) shows the overall amplitude structure of the recording and reflects the effect of silence removal at the beginning and end of the signal.

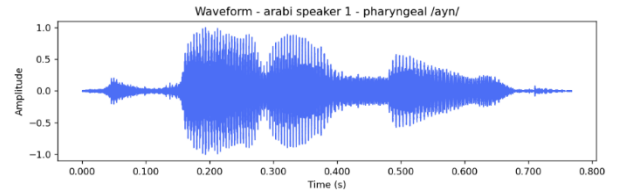


Figure 1: Waveform of arabi after preprocessing

Short-time energy further highlights temporal variations in speech intensity, as seen in *sadeeq* (Figure 3.2), where energy peaks correspond to more prominent speech segments within the recording. This representation helps identify regions of increased acoustic activity across the signal.

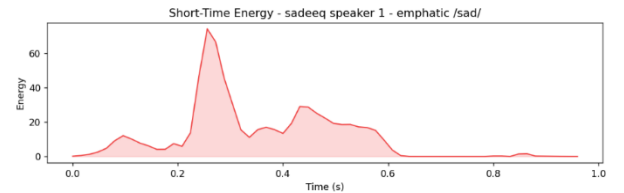


Figure 2: Short-time energy of sadeeq

Zero-crossing rate provides additional insight into the nature of the speech signal. In *thalatha* (Figure 3.3), higher zero-crossing activity is observed, which is consistent with the presence of the fricative /θ/, where rapid sign changes in the waveform are expected due to turbulent airflow.

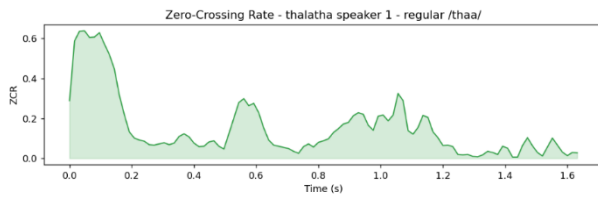


Figure 3: Zero-crossing rate of thalatha

3.3.3. Frequency-Domain Analysis

Frequency-domain analysis was performed using spectrograms and pitch contours to examine how spectral content varies across different Arabic speech sounds. The spectrogram represents how frequency energy is distributed over time and is particularly useful for observing how different consonant types influence the spectral structure of speech. In this study, uvular and pharyngeal sounds were of particular interest, as they tend to produce distinct spectral patterns due to their place of articulation.

This can be observed in the spectrogram of *qalam* (Figure 3.4), where the uvular consonant /q/ influences the distribution of energy in the higher frequency regions. Similarly, the spectrogram of *sadeeq* (Figure 3.5) shows the acoustic structure associated with the emphatic consonant /s/, which affects the surrounding spectral energy distribution.

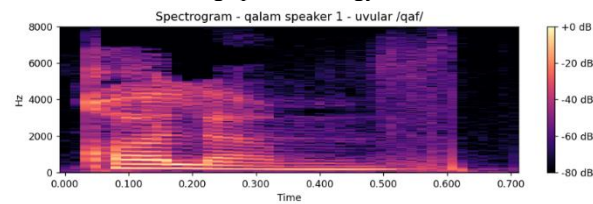


Figure 4: Spectrogram of qalam, showing uvular /q/ articulation effects

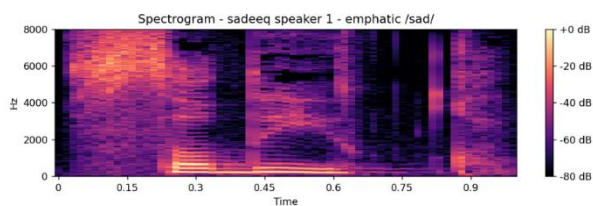


Figure 5: Spectrogram of sadeeq, showing emphatic /s/ acoustic pattern

Pitch contour analysis was used to estimate the fundamental frequency of voiced segments. Although pitch is not the primary feature for distinguishing consonant categories, it provides useful information about voicing and speech continuity. In this study, pitch was estimated using the probabilistic YIN method (librosa.pyin). The resulting pitch contour for *arabi* is shown in Figure 3.6, illustrating variations in fundamental frequency over the duration of the recording.

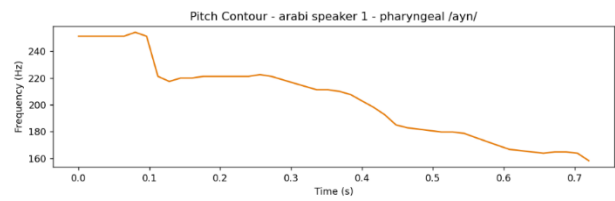


Figure 6: Pitch contour of arabi, showing fundamental frequency variation over time

All spectrograms were generated using a short-time Fourier transform (STFT) with a frame length of 512 samples and a hop length of 256 samples. Given the 16 kHz sampling rate, the spectral representation covers frequencies up to approximately 8 kHz due to the Nyquist limit.

3.3.4. Formant Analysis (Praat)

Formant analysis was conducted using Praat to obtain detailed phonetic information from the selected speech recordings. Praat was used to display the spectrogram and formant tracks (F1, F2, and F3), which were used to analyze vowel-related acoustic behavior in the speech signal.

Representative Praat outputs for *arabi*, *qalam*, and *sadeeq* are shown in Figures 3.7–3.9, illustrating examples from pharyngeal, uvular, and emphatic consonant contexts respectively.

The analysis was performed for all 10 selected words in the dataset. For each recording, formant values were extracted from a stable vowel region adjacent to the target consonant. This region was selected to minimize transitional effects and to obtain a more reliable estimate of vocal tract resonances. The first three formants (F1, F2, and F3) were recorded, as they provide key information about vowel quality and articulatory configuration.

Pharyngeal, uvular, and emphatic consonants may influence adjacent vowel formants due to their constriction and reconfiguration of the vocal tract. Therefore, comparing formant values across the selected words allows observation of potential acoustic variations associated with different articulation types. The complete set of extracted formant measurements for all 10 words is presented in Table 2.

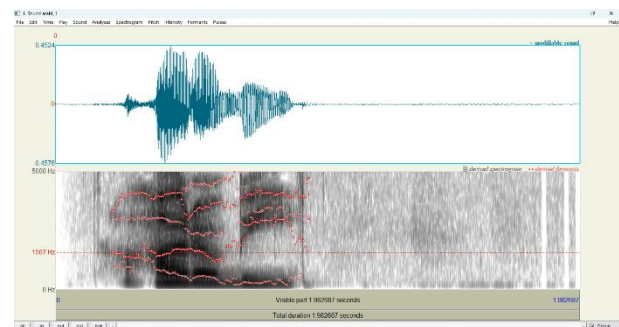


Figure 7: Praat formant analysis of arabi (F1, F2, F3 tracks)

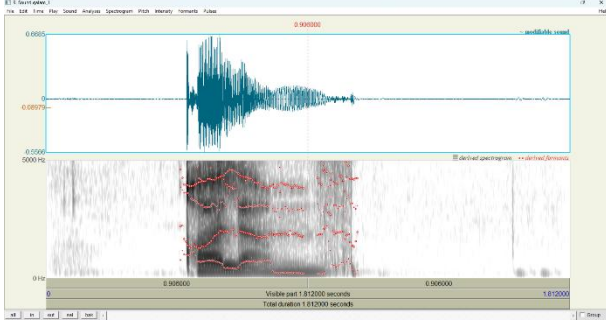


Figure 8: Praat formant analysis of qalam (F1, F2, F3 tracks)

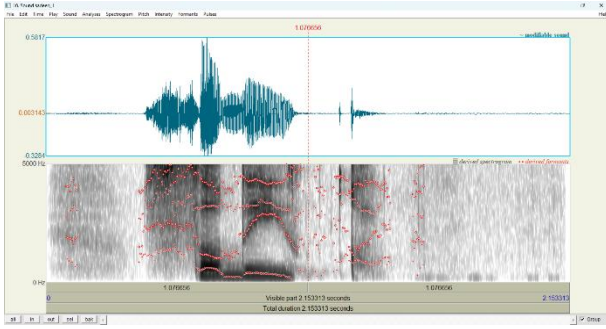


Figure 9: Praat formant analysis of sadeeq (F1, F2, F3 tracks)

Word	F1 (Hz)	F2 (Hz)	F3 (Hz)
Arabi	697.25	1953.45	2983.44
Daw	651.52	1608.32	3123.07
Ghurfa	796.80	1775.21	3076.67
Hadeka	783.94	2018.62	3077.96
Khalid	873.31	2220.39	3153.22
Qalam	767.14	1972.21	3204.68
Sadeeq	805.26	2051.17	3299.21
Tareeq	747.53	1989.88	3006.40
Thalatha	876.40	2044.52	3137.61
Zarf	839.38	2012.79	3156.99

Table 2: Formants (Praat Analysis)

3.3.5. Summary of Observations

The signal analysis showed that different Arabic word categories produced distinguishable acoustic patterns across the dataset. Waveform and short-time energy plots illustrated variations in amplitude and speech strength across recordings. Zero-crossing rate was useful for identifying high-frequency and noisy regions, particularly in fricative-containing words such as thalatha and sadeeq. Spectrograms provided a detailed representation of frequency content over time and highlighted differences in spectral structure across phonetic contexts. Pitch contours were used to identify voiced regions and observe general prosodic behavior, while Praat-based formant analysis provided additional insight into vocal tract resonances in vowel regions adjacent to target consonants.

Overall, combining Python-based signal analysis with Praat-based phonetic inspection provided a comprehensive acoustic description of the dataset prior to further analysis.

3.4. Feature extraction

Different acoustic features are employed in the proposed system of pronunciation assessment in order to capture the characteristics of speech production.

3.4.1. MFCC Features

Mel Frequency Cepstral Coefficients (MFCCs) were extracted from each audio file using the Librosa library. In total, 12 coefficients were used to extract spectral properties of the speech signal.

Additionally, temporal features such as Delta and Delta-Delta coefficients were extracted in order to obtain additional information regarding changes in the signal. Consequently, each recording is described by:

- 12 MFCC coefficients
- 12 Delta MFCC coefficients
- 12 Delta-Delta MFCC coefficients

for the total of 36 MFCC features.

3.4.2. Pitch

The pitch of each recording was estimated by using YIN pitch estimation algorithm and then used for measuring similarity between speakers.

3.4.3. Duration

The duration of each recording was measured in order to capture the difference in speaking speed and pronunciation timing.

3.4.4. Formants

Formants frequencies (F1, F2, and F3) were extracted using Linear Predictive Coding (LPC). Formants capture resonant properties of the vocal tract and describe articulation.

3.4.5. Additional Acoustic Features

Two additional acoustic features were calculated for each recording:

- Zero Crossing Rate (ZCR)
- Spectral Features (Spectral Centroid and Spectral Rolloff)

3.5. Scoring

Once all the acoustic features have been extracted from both the reference and test pronunciations, the pronunciation score can be obtained based on similarity comparison between the two pronunciations.

For MFCC features, the cosine similarity metric is used, while for Pitch, Duration, Formants, Zero Crossing Rate (ZCR) and Spectral features, normalized similarity metrics are used. Each feature generates a score value between 0 and 100, where a larger value means higher similarity with the reference pronunciation.

The final pronunciation score is derived based on a weighted sum of all feature scores. The weights are carefully chosen in such a way that they highlight the importance of those features

which are very relevant in terms of pronunciation, specifically MFCC and Formants features.

The final score lies between 0% to 100%, where:

- Higher scores imply higher pronunciation accuracy.
- Lower scores imply lower pronunciation accuracy or more deviation from the reference pronunciation.

3.6. Feedback

Once the final pronunciation score is calculated, the feedback is generated by the system to give users an idea about their pronunciation skills.

Feedback is based on predefined score intervals:

- Excellent Pronunciation: Very similar to the recording of the reference.
- Good Pronunciation: Sufficient pronunciation with some differences.
- Poor Pronunciation: Pronunciation with significant deviations from the reference pronunciation.

Besides the overall score, analysis of the individual features score may be helpful to detect pronunciation weaknesses. Thus, low Formant score may point to the difference in articulation, whereas low Duration score may indicate speaking rate differences.

This way, users will be able to easily evaluate their pronunciation skills.

4. Experiments and Results

4.1. Average Score per Word

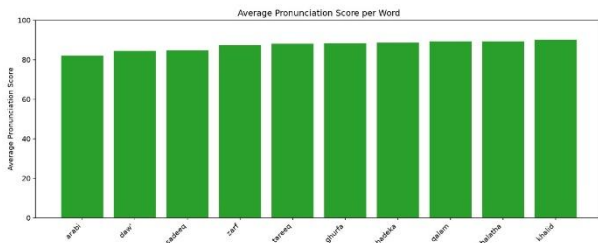


Figure 10: Average Pronunciation Score per Word

Figure 10 represents the average score for pronunciation of each Arabic word by all the speakers. It can be noticed that the average score for the proposed pronunciation assessment system was very high for all words that were assessed, reaching from 82% to 90%.

The word "arabi" had the smallest average score (about 82%). Thus, it seems that this word had the most variation among speakers. This may be related to different pronunciation style, speaking speed, or acoustic characteristics that affected the extracted features.

At the same time, "khalid", "thalatha", and "qalam" had the largest average scores reaching about 90%. Therefore, it may be concluded that there was more consistency in the pronunciation of these words among speakers and the extracted acoustic features were highly similar to those of reference recordings.

It should be noted that there is not much variation between the words which testifies to the efficiency of the suggested system. Despite the variations in the gender, pitch, speaking speed, and recording conditions, the system provided stable pronunciation assessment.

In conclusion, the findings prove that the proposed system is capable of evaluating the pronunciation of Arabic speech and distinguishing it from other variations of the pronunciation, thus making it a good system for educational purposes.

4.2. Feedback Distribution

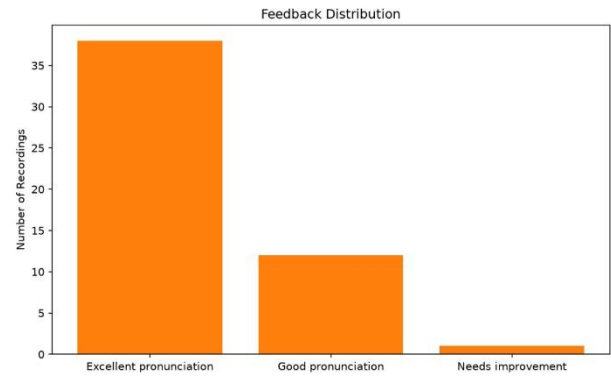


Figure 11: Feedback Distribution

Figure 11 illustrates the distribution of the pronunciation feedback provided by the developed system. The largest group of recordings (38 out of 51) was characterized as **Excellent Pronunciation, which means that most participants produced the pronunciation very close to the target pronunciation.

In addition, there was a smaller number of recordings (12 recordings) marked as Good Pronunciation. It should be noted that the participants' pronunciation contained some acoustic discrepancies compared to the target pronunciation like variations in pitch, speed of speaking, and articulation, but this pronunciation was still rated as acceptable.

Finally, only one recording was identified as Needs Improvement, which means that this participant produced the pronunciation with a significant difference from the target pronunciation.

Thus, the feedback distribution described above implies that the pronunciation assessment system developed by the researcher provides meaningful feedback. The large share of Excellent and Good categories can be explained by both the high quality of the recorded data and good functionality of the system.

4.3. Correct vs Incorrect Pronunciation Experiment

To test the effectiveness of the system for detecting pronunciation errors, another experiment was conducted using a deliberately mispronounced audio file. The reference audio file (arabi_1.wav) was compared not only to those made by other speakers who had pronounced the word correctly but also to the deliberately mispronounced file (arabi_wrong.wav).

The mispronounced file was created using the deliberate distortion of pronunciation of the target word to create a typical pronunciation error. The scores achieved were compared to the scores of the correctly pronounced files.

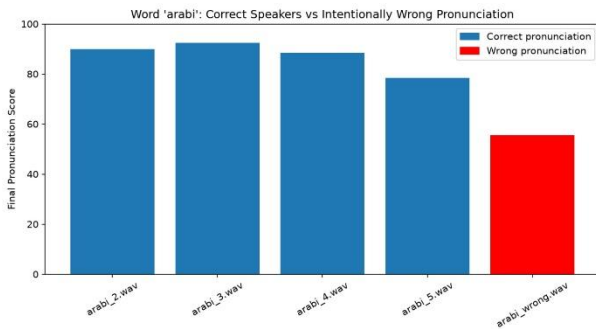


Figure 12: Correct Speaker vs Intentionally Wrong Pronunciation for 'arabi'

Figure 5.4 shows a comparison between the pronunciation scores for the correctly pronounced recordings and the purposely mispronounced recording of the word "arabi". The scores for the correctly pronounced recordings were relatively high ranging from around 79% to 92%, showing significant resemblance to the reference pronunciation.

On the other hand, the purposely mispronounced recording received significantly lower scores, reaching around 56%. The low scores show how sensitive the pronunciation assessment system proposed here is to any pronunciation mistakes.

It is clear through the results that the pronunciation characteristics can be captured by the acoustic features extracted, such as MFCC, pitch, duration, and formants. Therefore, the system will be able to recognize any departure from the reference pronunciation and give valid feedback on pronunciation.

This experiment proves the ability to detect pronunciation mistakes which is one of the main goals of this project.

4.3.1. Feature-Level Analysis

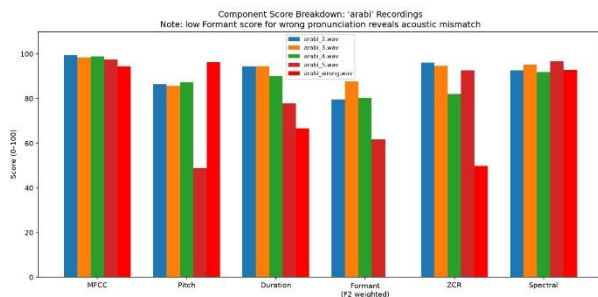


Figure 13: Feature-Level Comparison Between Correct and Incorrect Pronunciation of the Word 'arabi'

In order to perform an in-depth analysis of the operation of the pronunciation assessment system, a component-level analysis was carried out for the word "arabi". Figure 13 shows the results of the evaluation performed on each acoustic

component, which includes MFCC, Pitch, Duration, Formants, Zero Crossing Rate (ZCR), and Spectral components.

The correctly pronounced utterances scored highly in all feature categories, which reflects the high level of acoustic resemblance to the reference pronunciation. On the other hand, the purposely mispronounced utterance displayed lower scores in several feature components, namely Formant, ZCR, and Duration.

The lowest score is observed in the Formant component, and since formants reflect the acoustic properties of vowels and their articulation, it can be concluded that the acoustic structure of the mispronounced word differs greatly from the reference pronunciation.

In the same vein, the ZCR and Duration scores showed significant decrease when the pronunciation was wrong, indicating different speech timings and characteristics. Though the MFCC and Spectral scores still exhibited high values, the total reduction in these feature scores led to a pronounced decrease in the pronunciation score.

From these findings, it is evident that the suggested pronunciation scoring system is advantageous when using several acoustic features instead of one measurement. This analysis at the feature level validates that the system is able to detect pronunciation error through acoustic misalignment with respect to the recorded pronunciation.

4.4. Overall Pronunciation Results

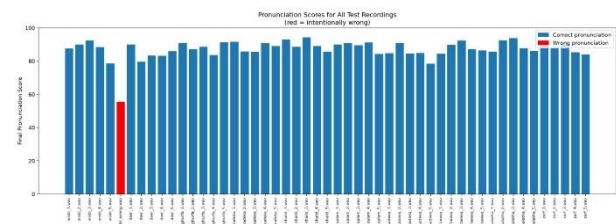


Figure 14: Final Pronunciation Scores for all Test Recordings

Figure 14 illustrates the pronunciation score results for all test recordings which were used in the dataset of the evaluation phase. Most of the recordings scored between 80% and 95%, showing a high similarity to the related reference pronunciation. It is evident that the proposed system is able to assess the pronunciation of different speakers with consistent results.

Although there are natural differences in the pitch, speaking rate, and articulation manner of different speakers, the scores do not vary much. Therefore, the selection of acoustic features in the developed system is successful in capturing the main features of the pronunciation, and it is robust to speaker variation.

The deliberately mispronounced recording, which is colored in red, had a score considerably lower than the others. It is evident from the results that the system is able to distinguish between the normal pronunciation variations and pronunciation mistakes.

Therefore, the obtained results show that the proposed pronunciation assessment system is accurate and robust to speaker variations and also is able to detect the pronunciation mistakes.

4.5. Live Demonstration Results

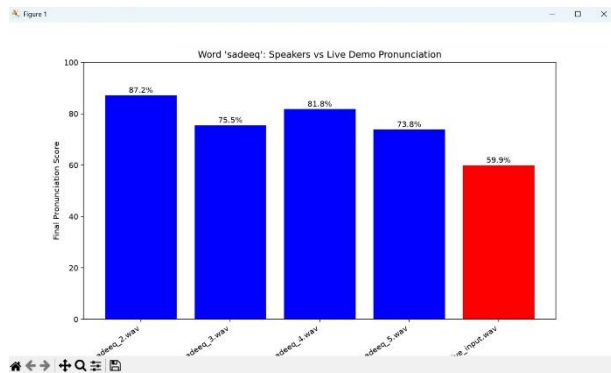


Figure 15: Live Pronunciation Assessment for Word 'sadeeq'

Results of the live demo for the word "sadeeq" are presented in Figure 15. The live pronunciation demo was able to get a score of about 59.9%, which is lower than the scores of the pre-recorded speakers. While the speaker recording demos got scores that range from approximately 73.8% to 87.2%, the live demo had a more varied pronunciation compared to the reference pronunciation.

This is proof of how the system can analyze live speech and measure pronunciation accuracy. The lower score might have been due to the speaking style, recording environment, microphone, or pronunciation accuracy in the live demo.

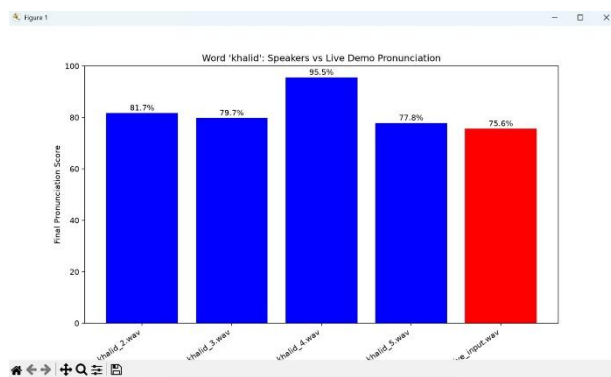


Figure 16: Live Pronunciation Assessment for Word 'Khalid'

Figure 16 demonstrates the live demonstration scores of the word "khalid". The live pronunciation managed to have a score of about 75.6%, which is roughly similar to a number of speaker recordings from the prerecorded database. Although another speaker managed to get a better score of about 95.5%, the live recording still falls under the scope of acceptable pronunciation.

It can be seen from the results that the suggested system is capable of evaluating live pronunciation and generating scores accordingly.

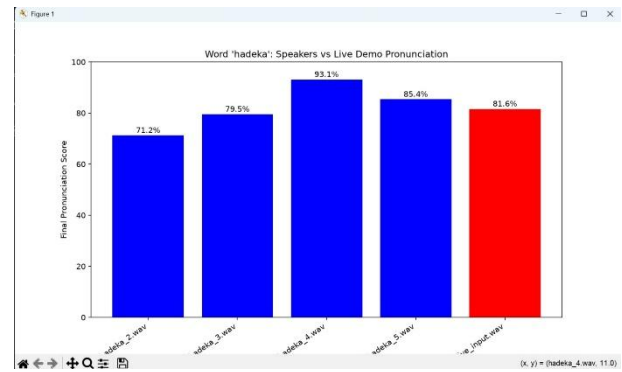


Figure 17: Live Pronunciation Assessment for Word 'hadeka'

Results of the live evaluation test for the word "hadeka" are presented in Figure 17. From the live audio sample, it was found that the degree of accuracy was around 81.6% that is very similar to the ones found from the prerecorded samples. This result implies that the live audio sample is acoustically similar to the reference sample and it was considered to be the correct pronunciation.

Moreover, from the results, it could be seen how the proposed system is used practically in the online system since it was clearly proven that the system can analyze the live sample as well as compare it to the reference sample.

5. Team member contribution

The project was completed collaboratively, with responsibilities divided between the two team members. Raghad Jamhour was responsible for dataset collection and organization, audio preprocessing, signal analysis, and writing the Abstract, Introduction, Background, Methodology (3.1-3.3), and Conclusion sections of the report. Raghad also assisted with the implementation of the system code, experiment execution, and analysis of the experimental results. Woroud Amjad was responsible for implementing the feature extraction, similarity comparison, scoring, and feedback generation modules. Woroud also conducted the experimental evaluation, generated the result figures, and wrote the Feature Extraction, Scoring, Feedback, Experiments and Results sections.

6. Conclusion

This project presented an automatic Arabic pronunciation assessment system based on speech signal processing techniques. The objective of developing a lightweight system capable of evaluating Arabic pronunciation and providing meaningful feedback was successfully achieved. The system combined multiple acoustic features, including MFCCs, pitch, duration, formants, zero-crossing rate, and spectral features, to compare a test pronunciation with a reference recording.

Experimental results showed that the proposed system produced consistent pronunciation scores across different speakers while successfully identifying intentionally incorrect pronunciations. The feature-level analysis demonstrated that

combining several acoustic features provides a more reliable assessment than relying on a single feature alone.

One limitation of the current system is the relatively small dataset and the use of word-level rather than phoneme-level analysis, which may limit generalization to a wider range of speakers and pronunciations. Future work could include expanding the dataset, incorporating phoneme or vowel segmentation for more detailed analysis, and investigating machine learning or deep learning approaches to further improve pronunciation assessment accuracy.

Overall, the project demonstrates that classical speech signal processing techniques can provide an effective and interpretable solution for automatic Arabic pronunciation assessment without requiring complex deep learning models.

7. References

- [1] S. B. Davis and P. Mermelstein, "Comparison of parametric representations for monosyllabic word recognition in continuously spoken sentences," *IEEE Transactions on Acoustics, Speech, and Signal Processing*, vol. 28, no. 4, pp. 357–366, 1980.
- [2] M. Eskenazi, "An overview of spoken language technology for education," *Speech Communication*, vol. 51, no. 10, pp. 832–844, 2009.
- [3] Automatic Pronunciation Assessment and Feedback for Arabic Learners: A Review, *International Journal of Asian Language Processing*, 2024. [Online]. Available: <https://www.worldscientific.com/doi/10.1142/S2717554524300019>
- [4] S. S. Çalık, A. Küçükmanisa, and Z. H. Kilimci, "A novel framework for mispronunciation detection of Arabic phonemes using audio-oriented transformer models," *Applied Acoustics*, vol. 215, p. 109711, 2024.
- [5] Boersma, P. and Weenink, D., "Praat: doing phonetics by computer," [Computer program], Version 6.x, retrieved from <http://www.praat.org/>
- [6] OpenAI, "ChatGPT," <https://chat.openai.com/>
- [7] Anthropic, "Claude," <https://claude.ai/>